Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 6)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The midpoint of the line segment joining $(4, -2)$ and $(-2, 4)$ is: [1]

(A) $(1, 1)$ (B) $(2, 2)$ (C) $(3, 3)$ (D) $(0, 0)$

Q2. The line represented by $\frac{x}{4} + \frac{y}{6} = 1$, intersects x-axis and y-axis respectively at P and Q. The coordinates of the mid-point of line segment PQ are : [1]

(A) $(2, 3)$ (B) $(3, 2)$ (C) $(2, 0)$ (D) $(0, 3)$

Q3. If the mid-point of the line segment joining the points $(a, 4)$ and $(2, 2b)$ is $(2, 6)$, then the value of $(a + b)$ is given by : [1]

(A) 6 (B) 7 (C) 8 (D) 16

Q4. The mid-point of the line segment joining the points $P(-4, 5)$ and $Q(4, 6)$ lies on : [1]

(A) x-axis (B) y-axis (C) origin (D) neither x-axis nor y-axis

Q5. If the points $A(5, 4)$ and $B(x, 6)$ are on a circle with centre $(3, 4)$, then the value of x is : [1]

(A) 6 (B) 7 (C) 3 (D) 1

SECTION B: Short Answer Type Questions (6 Marks)

Q6. Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$. [2]

Q7. Find the coordinates of the point which divides the line segment joining the points $(4, -3)$ and $(8, 5)$ in the ratio 3 : 1 internally. [2]

Q8. Find the coordinates of the point which divides the line segment joining the points $(-1, 7)$ and $(4, -3)$ in the ratio 2 : 3 internally. [2]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q9. Find the ratio in which the point $P(-1, y)$ lying on the line segment joining points $A(-3, 10)$ and $B(6, -8)$ divides it. Also find the value of y. **[5]**

Q10. Competency Case Study: A smart park security system tracks a drone patrol path between camera points $A(2, 3)$ and $B(8, 7)$ on a grid layout. **[4]**

(i) Find the straight distance between cameras A and B.

(ii) Find the coordinates of the central charging hub placed at the midpoint of AB.

(iii) If a sensor station C divides the patrol path AB in the ratio $1 : 2$, find the coordinates of C.

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